

🔥 XAI: Understanding Black-Box Predictions via Influence Functions

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

What is Your Data Worth? Equitable Valuation of Data

Amirata Ghorbani, James Zou — ICML 2019

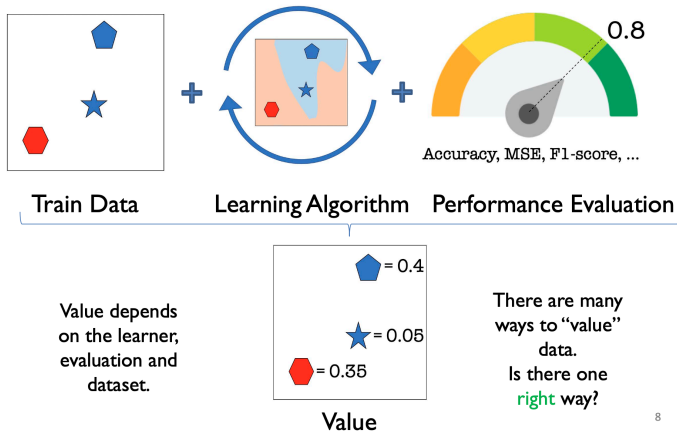
(Ghorbani and Zou 2019)

🍲 The Data Valuation Problem -----

If data is fuel, we need a principled way to measure its value. Stakeholders have different priorities:

- **ML Engineers:** Assess heterogeneous sources and data quality.
- **Data Vendors:** Determine fair pricing for buying and selling data.
- **Individuals/Data Producers:** Understand compensation or credit for contributed data.

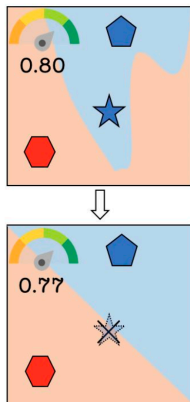
🍲 Ingredients of Data Value -----



8

🍷 Leave One Out Method -----

Example: value (★) = $0.80 - 0.77 = 0.03$



🍷 Desirable Properties of Valuation -----

A fair valuation method must satisfy three fundamental axioms from cooperative game theory:

- ① **Null Element:** If adding a point to *any* subset of training data never changes the model's performance, its value is exactly 0.
- ② **Symmetry:** If adding point A or point B to any subset always results in the exact same performance change, they must receive the same value ($Value(A) = Value(B)$).
- ③ **Linearity:** If the performance metric is a sum of performances on individual tasks, the data value must reflect the sum of values for those individual tasks.

The Data Shapley Value -----

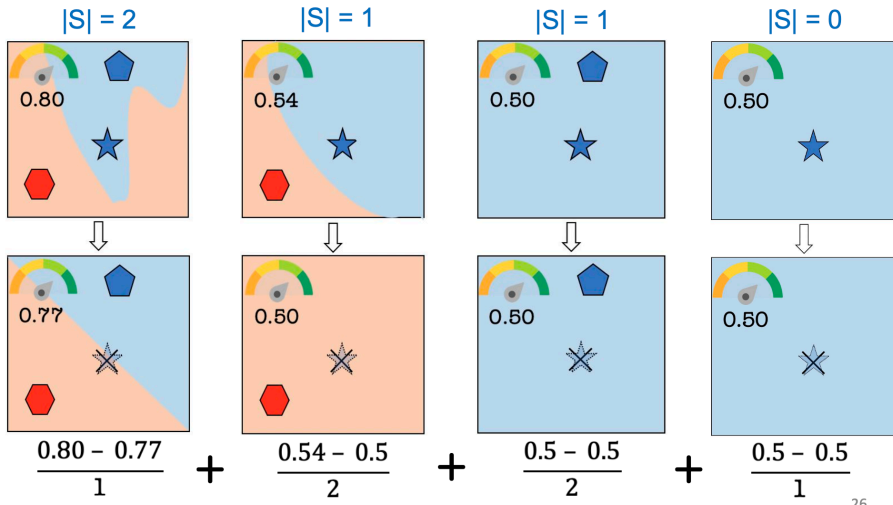
The only data value mathematically guaranteed to satisfy these three properties is based on marginal contributions across all possible subset sizes.

$$Value(data\ k) = \sum_{S \subseteq D \setminus \{k\}} \frac{performance(S \cup \{k\}) - performance(S)}{\binom{n-1}{|S|}}$$

- Evaluates the expected contribution to all possible sizes of train data samples.
- Directly applies Lloyd Shapley's cooperative game theory to individual ML data points.

🥥 The Data Shapley Value Example -----

Example: value (★) = 0.05



🍷 Efficient Approximation: TMC-Shapley -----

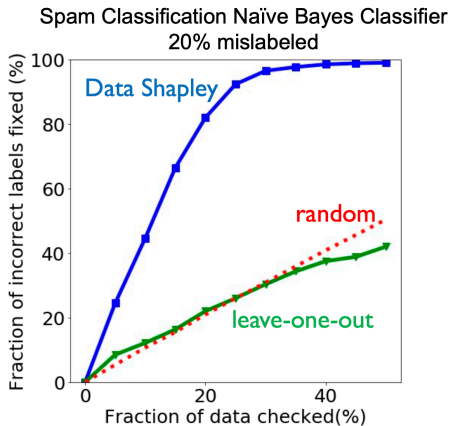
Exact calculation is computationally infeasible for realistic datasets because it grows exponentially.

Truncated Monte-Carlo (TMC) Approximation:

- 1 Sample a random permutation of the training data.
- 2 Add one point at a time based on the sampled permutation.
- 3 Re-train the model to capture the marginal increase in performance.
- 4 **Truncate** the calculation when performance saturates to save computational resources.

🍲 Application 1: Identifying Low-Quality Data -----

Result: Ordering points from most negative Shapley value to positive allows for rapid dataset cleaning. Examining just ~30% of the data uncovers nearly all mislabeled examples.



🍲 Application 2: Identifying Essential Data -----

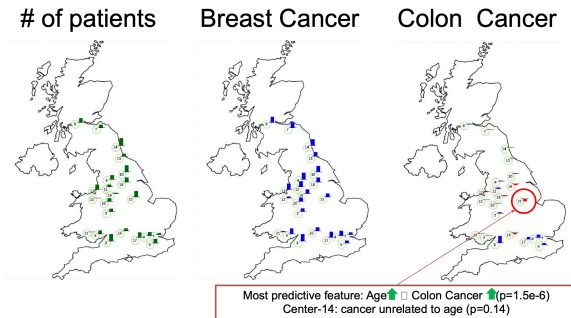
Case Study: UK Biobank

- Dataset spanning 500,000 individuals across 22 centers in the UK.
- Centers were evaluated as singular “data sources” for predicting Breast and Colon Cancer.

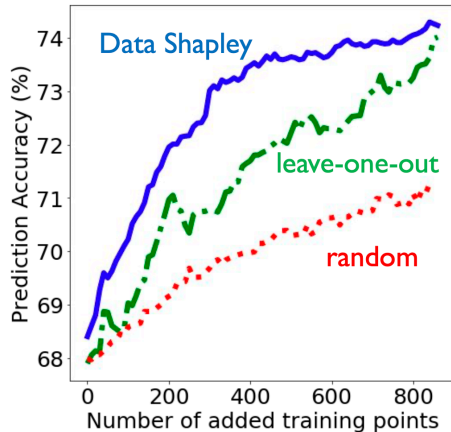
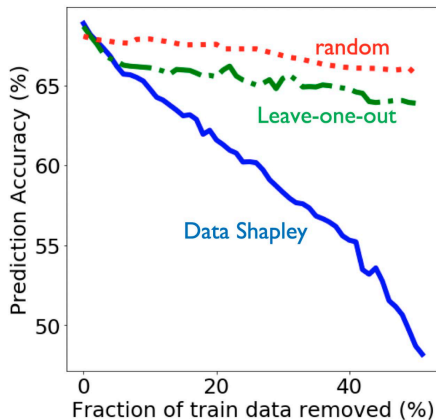
🍷 Application 2: Identifying Essential Data -----

Insights from Shapley:

- For Colon Cancer, a massive center received a *negative* Shapley value.
- Investigation revealed the model heavily relied on *Age* as a predictive feature, but the specific anomalous center exhibited colon cancer rates entirely independent of age.



🍵 Application 2: Identifying Essential Data -----



Application 3: Domain Adaptation -----

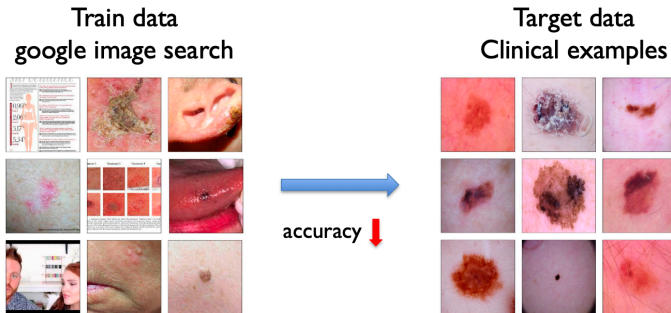
What if the training data differs in quality, distribution, or class balance from the test environment?

Shapley Adaptation Strategy:

- 1 Compute Shapley values relative to the target test distribution.
- 2 Remove data with negative values.
- 3 Reweight the remaining training data based on relative positive weights.

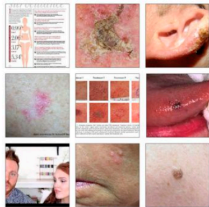
🍵 Domain Adaptation Results (Skin Lesion) -----

Training on noisy Google Image Search data to test on the clean, clinical HAM10000 dataset.
Applying Shapley weights improved classification by $\approx 25\%$.



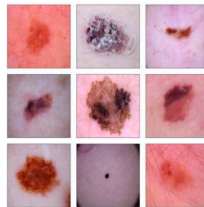
🍵 Domain Adaptation Results (Skin Lesion) -----

Train Data: Google Images



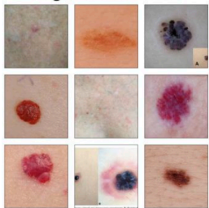
29.6%

Test Data: HAM10000



≈ 25% ↑

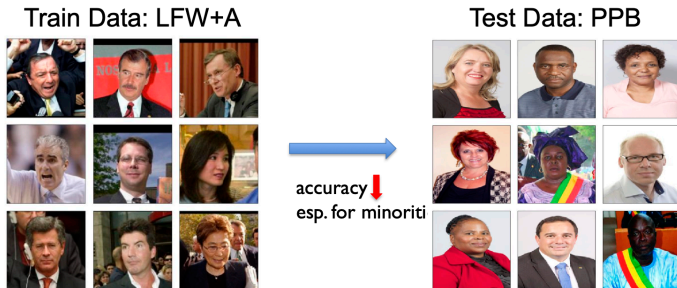
High Value Data



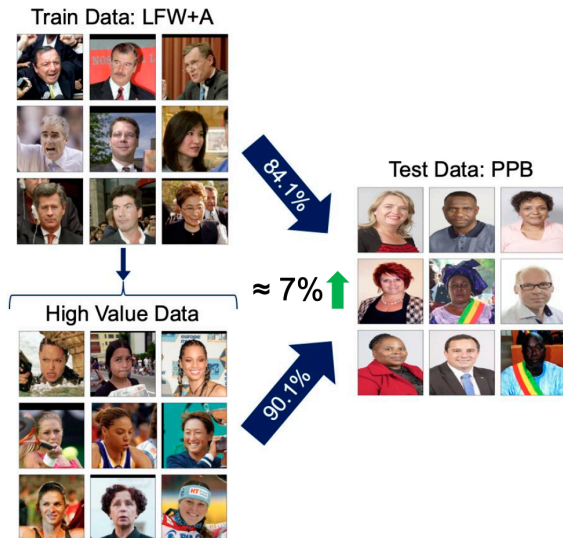
37.8%

🍷 Domain Adaptation Results (Gender Detection) -----

Models trained on LFW+A (heavily biased toward white males) fail on balanced datasets like PPB. Shapley values easily identify out-of-distribution elements; adaptation increases accuracy on underrepresented minorities by $\approx 7\%$.



🍲 Domain Adaptation Results (Gender Detection) -----



Summary -----

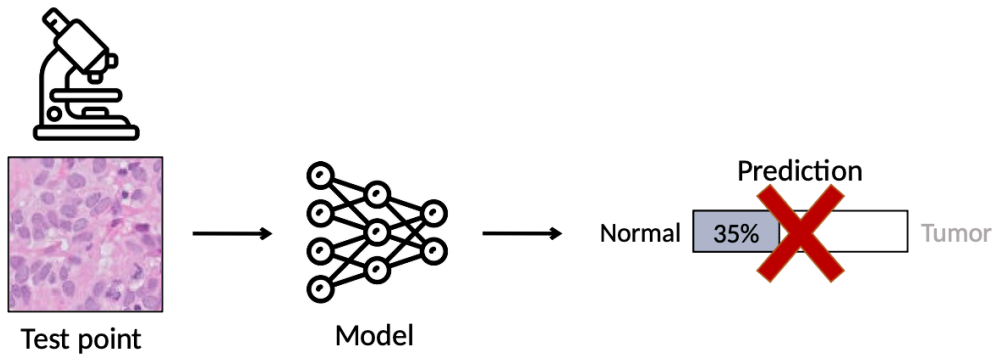
- Data Shapley provides an equitable, axiom-backed framework to quantify the value of individual ML data points.
- It moves beyond LOO limitations by measuring expected contributions over all possible subsets.
- Applications span data cleaning, debugging, valuation pricing, and addressing distributional shifts.

Understanding black-box predictions via influence functions

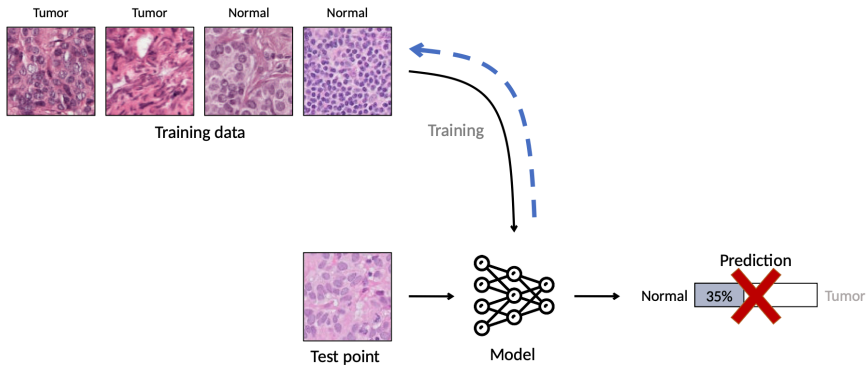
Pang Wei Koh, Percy Liang — ICML 2017

(Koh and Liang 2017)

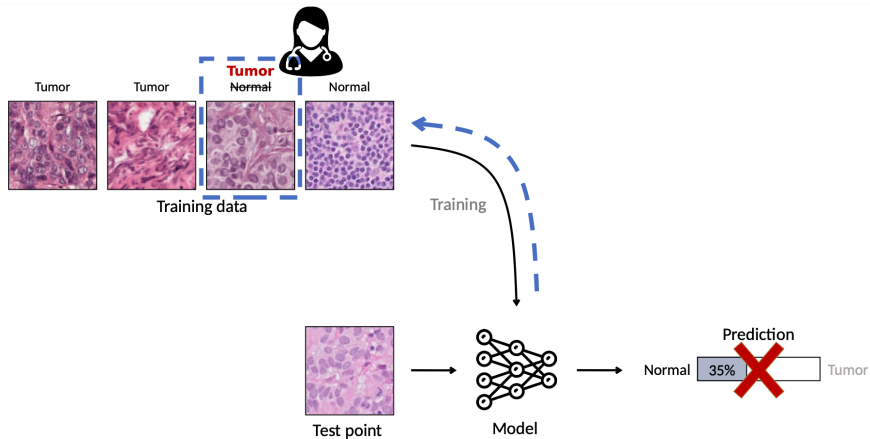
🍷 Dataset Debugging -----



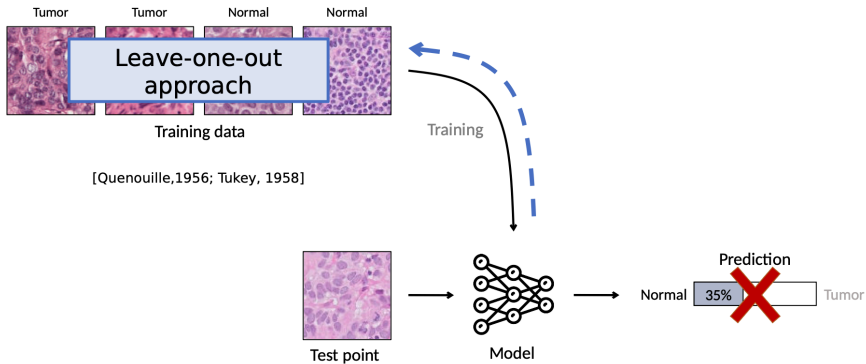
🍵 Dataset Debugging -----



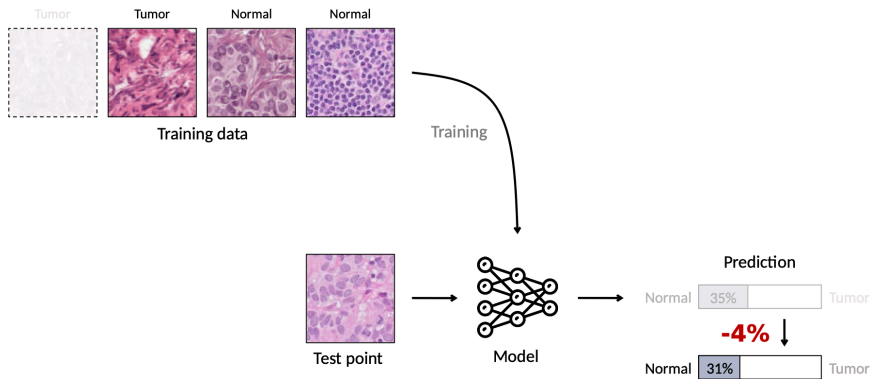
🍵 Dataset Debugging -----



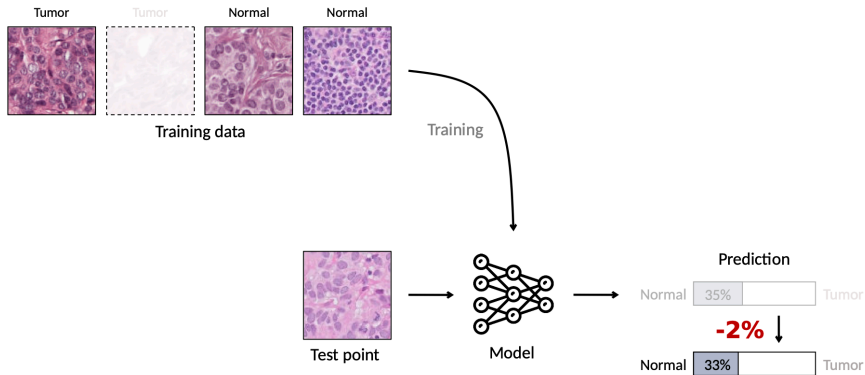
🍲 Dataset Debugging -----



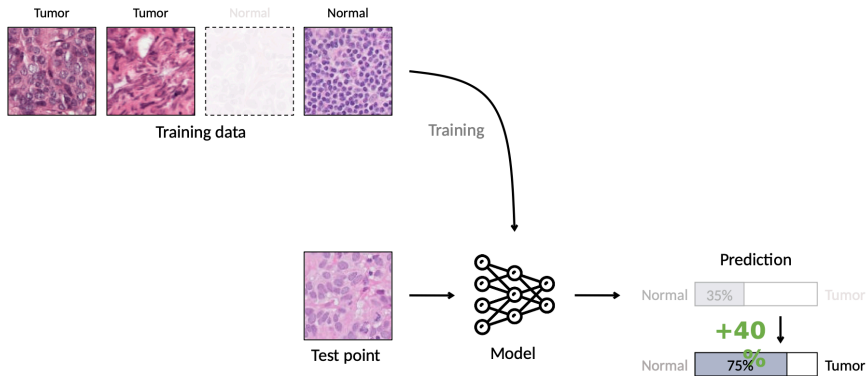
🍵 Dataset Debugging -----



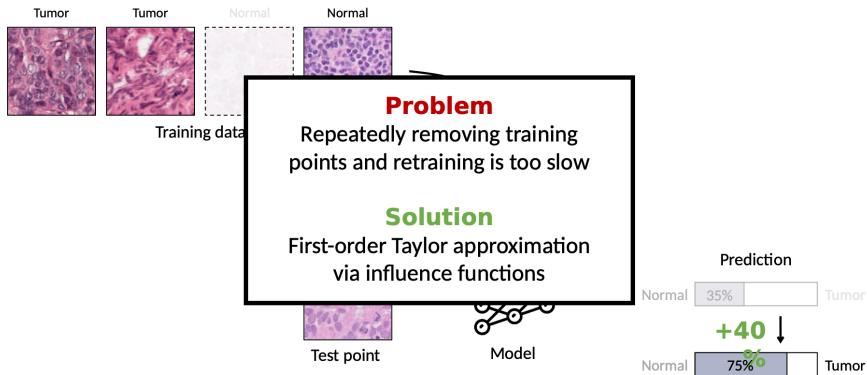
🍷 Dataset Debugging -----



🍵 Dataset Debugging -----



🍲 Dataset Debugging -----



🍷 The Influence Function Approximation -----

Change in loss on after removing

Model's representation of

Effect of other training points

Model's representation of

$$\ell(z_{test}, \hat{\theta}_{-z_{train}}) - \ell(z_{test}, \hat{\theta}) \approx \nabla_{\theta} \ell(z_{test}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_{\theta} \ell(z_{train}, \hat{\theta})$$

Gradient of loss on

Inverse of the Hessian

Gradient of loss on

The diagram illustrates the Influence Function Approximation equation. The equation is: $\ell(z_{test}, \hat{\theta}_{-z_{train}}) - \ell(z_{test}, \hat{\theta}) \approx \nabla_{\theta} \ell(z_{test}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_{\theta} \ell(z_{train}, \hat{\theta})$. Annotations include: a bracket above the left side labeled 'Change in loss on after removing'; a bracket above the first term of the right side labeled 'Model's representation of' with 'Gradient of loss on' below it; a bracket above the second term of the right side labeled 'Model's representation of' with 'Gradient of loss on' below it; a vertical arrow pointing down to the $H_{\hat{\theta}}^{-1}$ term labeled 'Effect of other training points'; and a vertical arrow pointing up to the $H_{\hat{\theta}}^{-1}$ term labeled 'Inverse of the Hessian'.

Doesn't require retraining!

Interpretation: Model Representations -----

$$\ell(z_{\text{test}}, \hat{\theta}_{-z_{\text{train}}}) - \ell(z_{\text{test}}, \hat{\theta}) \approx \nabla_{\theta} \ell(z_{\text{test}}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_{\theta} \ell(z_{\text{train}}, \hat{\theta})$$

Gradient = model's representation

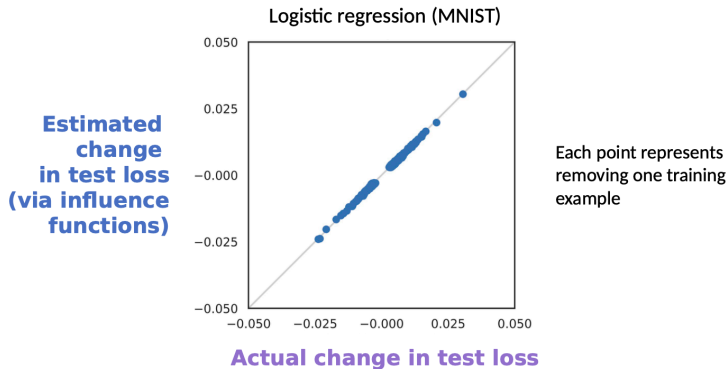
$\nabla_{\theta} \ell(z_{\text{test}}, \hat{\theta})$ encodes how the model "sees"
 z_{test}

$H_{\hat{\theta}}^{-1}$ = effect of other training points

Accounts for the curvature induced by the
full training set

High influence $\Rightarrow$ test and train have **similar model representations**

🍷 Removing Single Points: Validation -----



Logistic regression (MNIST): each point = removing one training example. Influence function estimate tracks actual LOO change closely.

Robustness

- Training set biases [Ren et al., 2018]
- Inference reliability [Broderick et al., 2021]
- Cross-validation [Stephenson et al., 2020]
- Memorization [Feldman, 2019]

Applications

- Data distillation [Wang et al., 2020]
- Data valuation [Jia et al., 2019]
- Active learning [Gudovskiy et al., 2020]
- Data debugging [Guo et al., 2021]

Fairness & Security

- Algorithmic bias [Verma et al., 2021]
- Data labor [Arrieta-Ibarra, 2018]
- Privacy [Shokri et al., 2021]
- Data poisoning [Chen et al., 2017]

🍷 Limitations -----

Key assumptions required

- What assumptions on models are needed to calculate influence functions?
- Can you calculate influence functions for a neural network?

$$\hat{\theta}_{\epsilon, z} \stackrel{\text{def}}{=} \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n L(z_i, \theta) + \epsilon L(z, \theta)$$

- **Non-convexity:** derivation assumes a convex loss landscape
- **Non-convergence:** derivation assumes the optimizer has converged

"If Influence Functions are the Answer, Then What is the Question?"

Bae, Ng, Lo, Ghassemi & Grosse

🍷 Summary -----

- **Link** model behavior to training data
- **Efficient** to calculate — no retraining needed
- **Many applications:** debugging, fairness, security, privacy
- **Limitations** when applying to complex models
 - ▶ Non-convexity
 - ▶ Non-convergence



Thank You!

References ----- |

- Ghorbani, Amirata, and James Zou. 2019. “Data Shapley: Equitable Valuation of Data for Machine Learning.” In *Proceedings of the 36th International Conference on Machine Learning*, edited by Kamalika Chaudhuri and Ruslan Salakhutdinov, vol. 97. Proceedings of Machine Learning Research. PMLR.
<https://proceedings.mlr.press/v97/ghorbani19c.html>.
- Koh, Pang Wei, and Percy Liang. 2017. “Understanding Black-Box Predictions via Influence Functions.” *International Conference on Machine Learning*, 1885–94.